

Total No. of Questions : 8]

SEAT No. :

PE-5805

[Total No. of Pages : 2

[6584]-455

B.E. (Computer Engineering) (Honors)

INTERNET OF THINGS AND EMBEDDED SECURITY

(2019 Pattern) (Semester - VII) (410401)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume Suitable data if necessary.

Q1) a) Explain an IoT Security CONOPS document, network and security integration? **[8]**

b) Explain in details system security verification and validation (V&V) model. **[9]**

OR

Q2) a) Explain Operations and Maintenance in IoT security with focus on identity and security monitoring. **[8]**

b) Explain the secure IoT system implementation life cycle **[9]**

Q3) a) Explain the role of cryptography in ensuring security goals (CIAAN) in IoT systems **[9]**

b) Explain the cryptographic controls in IoT communication protocols such as ZigBee, Bluetooth, and NFC **[9]**

OR

Q4) a) Explain the cryptographic mechanisms used in IoT messaging protocols like MQTT, CoAP, DDS, and REST. **[9]**

b) Explain symmetric and asymmetric encryption with suitable IoT use cases. **[9]**

P.T.O.

Q5) a) Explain secure bootstrap, credential provisioning, and account monitoring in IoT IAM. **[8]**

b) Explain the identity lifecycle in IoT with an example of device onboarding and sunset. **[9]**

OR

Q6) a) Write short notes on authentication credentials used in IoT **[8]**

i) passwords

ii) symmetric keys

iii) certificates

iv) biometrics.

b) Describe the IAM infrastructure for IoT, focusing on 802.1X and PKI integration. **[9]**

Q7) a) Compare federated identity and global web identity with examples. **[9]**

b) Describe identity portrayal and its importance in identity management. **[9]**

OR

Q8) a) Compare local identity vs. network identity models. **[9]**

b) Design a secure IoT identity model for smart home or smart healthcare systems. **[9]**

